DA-1

Question 1:

Code:

```c
#include <stdio.h>

int main(){
    int a;
    scanf("%d",&a);
    int arr[a];
    for (int i=0;i<a;i++){
        scanf("%d",&arr[i]);
    }
    int arr1[a];
    for (int i=0;i<a;i++){
        arr1[i]=arr[i];
    }
    printf("\n");
    printf("Original Array:\n");
    for (int i=0;i<a;i++){
        printf("%d ",arr[i]);
    }
    int j=0;
    for (int i=0;i<a;i++){

        if (arr1[i]%2!=0){
            arr[j]=arr1[i];
            j+=1;
        }
        else{
            continue;
        }
    }
    for (int i=0;i<a;i++){
        if (arr1[i]%2==0){
            arr[j]=arr1[i];
            j+=1;
        }
        else{
            continue;
        }
    }
    printf("\n");
    printf("Final Sorted Array:\n");
    for (int i=0;i<a;i++){
        printf("%d ",arr[i]);
    }

}
```

Output:

```
6
1
2
3
4
5
6

Original Array:
1 2 3 4 5 6
Final Sorted Array:
1 3 5 2 4 6 %
harshitsingh@Harshits-MacBook-Air DA_1 % cd "/Users/harshitsingh/Documents/C/DA_1/" && gc
c main.c -o main && "/Users/harshitsingh/Documents/C/DA_1/"main
4
20
30
41
43

Original Array:
20 30 41 43
Final Sorted Array:
41 43 20 30 %
harshitsingh@Harshits-MacBook-Air DA_1 %
```

Question 2:

```c
#include <stdio.h>

int main(){
    int a,b;
    scanf("%d %d",&a,&b);
    int arr[a];
    int arr1[b];
    int arrf[a+b];
    for (int i=0;i<a;i++){
        scanf("%d",&arr[i]);
    }
    for (int i=0;i<b;i++){
        scanf("%d",&arr1[i]);
    }

    for (int i=0;i<a-1;i++){
        for (int j=0;j<a-i-1;j++){
            if(arr[j]>arr[j+1]){
                int temp=arr[j];
                arr[j]=arr[j+1];
                arr[j+1]=temp;
            }
        }
    }
    for (int i=0;i<b-1;i++){
        for (int j=0;j<b-i-1;j++){
```

```
            if(arr1[j]>arr1[j+1]){
                int temp=arr1[j];
                arr1[j]=arr1[j+1];
                arr1[j+1]=temp;
            }
        }
    }
    printf("Array 1:\n");
    for (int i=0;i<a;i++){
        printf("%d ",arr[i]);
    }
    printf("\nArray 2:\n");
    for (int i=0;i<b;i++){
        printf("%d ",arr1[i]);
    }
    for (int i=0;i<a;i++){
        arrf[i]=arr[i];
    }

    for (int i=a;i<a+b;i++){
        arrf[i]=arr1[i-a];
    }
    printf("\nFinal Sorted Array:\n");
    for (int i=0;i<a+b;i++){
        printf("%d ",arrf[i]);
    }
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air Question_6 % cd "/Users/harshitsingh/Documents/DA_CSE/Question_2/" && gcc main.c
-o main && "/Users/harshitsingh/Documents/DA_CSE/Question_2/"main
5 4
5
1
2
7
9
11
13
4
5
Array 1:
1 2 5 7 9
Array 2:
4 5 11 13
Final Sorted Array:
1 2 5 7 9 4 5 11 13 %
harshitsingh@Harshits-MacBook-Air Question_2 % cd "/Users/harshitsingh/Documents/DA_CSE/Question_2/" && gcc main.c -
o main && "/Users/harshitsingh/Documents/DA_CSE/Question_2/"main
3 2
3
2
1
6
7
Array 1:
1 2 3
Array 2:
6 7
Final Sorted Array:
1 2 3 6 7 %
harshitsingh@Harshits-MacBook-Air Question_2 %
```

Question 3:

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main (){
    char str[100], word[100], twoD[10][30];
    int i = 0, j = 0, k = 0, len1 = 0, len2 = 0;
    printf ("Enter the string:\n");
    gets (str);
    printf ("Enter the word to be removed:\n");
    gets (word);
    for (i = 0; str[i] != '\0'; i++){
    if (str[i] == ' '){
            twoD[k][j] = '\0';
            k ++;
            j = 0;
        }
        else{
            twoD[k][j] = str[i];
            j ++;
        }
    }
    twoD[k][j] = '\0';
    j = 0;
    for (i = 0; i < k + 1; i++){
        if (strcmp(twoD[i], word) == 0){
            twoD[i][j] = '\0';
        }
    }
    j = 0;
    for (i = 0; i < k + 1; i++){
        if (twoD[i][j] == '\0')
            continue;
        else
            printf ("%s ", twoD[i]);
    }
    printf ("\n");
    return 0;
}
```

Output:

```
My name is Xyz Xyz name
Enter the word to be removed:
name
My is Xyz Xyz
```

```
I love india
Enter the word to be removed:
I
love india
```

Question 4:

```c
#include <stdio.h>
#include <string.h>
int main(){
    char a[100];
    int count,f,m,l,l1;
    gets(a);
    for (int i=0;i<strlen(a);i++){
        if(a[i]==' '){
            count+=1;
        }
    }
    if (count==1){
        printf("%c.",a[0]);
        for (int i=0;i<strlen(a);i++){
        if(a[i]==' '){
            m=i+1;
        }
    }
    for(int i=m;i<strlen(a);i++){
        printf("%c",a[i]);
    }
    }
    else if(count==2){
        printf("%c.",a[0]);
        for(int i=0;i<strlen(a);i++){
            if(a[i]==' '){
                l=i+1;
                l1+=1;
                break;
            }
        }
        printf("%c.",a[l]);
        l1=0;
        for(int i=0;i<strlen(a);i++){
            if(a[i]==' '){
                l=i+1;
            }
        }
        for(int j=l;j<strlen(a);j++){
            printf("%c",a[j]);
        }

    }

}
```

Output:

```
1 warning generated.
warning: this program uses gets(), which is unsafe.
Harshit Singh Shekhawat
H.S.Shekhawat%
```

```
1 warning generated.
warning: this program uses gets(), which is unsafe.
Ram Kumar Sharma
R.K.Sharma%
```

Question 5:

```c
#include <stdio.h>
#include <string.h>

int main(){
    char str[100];
    gets(str);
    int count;
    for (int i=0;i<strlen(str);i++){
        if(str[i]==' '){
            count+=1;
        }
    }
    int arr[count+1];
    int count1,l;
    int j=0;
    for (int i=0;i<strlen(str);i++){
        if(str[i]!=' '){
            count1+=1;
        }
        else{
            l=i+1;
            arr[j]=count1;
            count1=0;
            j++;
        }


    }
    int count2;
    for (int i=l;i<strlen(str);i++){
        count2+=1;
    }
    arr[count]=count2;
    int max=arr[0];
    int max1;
```

```
    for(int i=0;i<count+1;i++){
        if (arr[i]>max){
            max=arr[i];
            max1=i;
        }
        }
    int count3,ls;
        for (int i=0;i<strlen(str);i++){
            if(str[i]==' '){
                count3+=1;
                if(count3==max1){
                    ls=i+1;
                    break;
                }
            }
        }
        for(int i=ls;i<strlen(str);i++){
            printf("%c",str[i]);
        if(str[i]==' '){
            break;
        }
     }
}
```

Output:

```
1 warning generated.
warning: this program uses gets(), which is unsafe.
my name abcdefghik and i love efg
abcdefghik %
```

Question 6:

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int my_strcmp(const char *str1, const char *str2) {
    return strcmp(str1, str2);
}

void my_swap(char **str1, char **str2) {
    char *temp = *str1;
    *str1 = *str2;
    *str2 = temp;
}

void sort(char *arr[], int n, int (*cmp)(const char *, const char *), void
(*swp)(char **, char **)) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (cmp(arr[j], arr[j + 1]) > 0) {
                swp(&arr[j], &arr[j + 1]);
            }
        }
    }
}

int main() {
    int n;

    printf("Enter the number of names: ");
    scanf("%d", &n);

    char *names[n];

    printf("Enter %d names:\n", n);
    for (int i = 0; i < n; i++) {
        char temp[100];
        scanf("%s", temp);
        names[i] = strdup(temp);
    }

    sort(names, n, my_strcmp, my_swap);


    printf("\nSorted names:\n");
    for (int i = 0; i < n; i++) {
        printf("%s\n", names[i]);
        free(names[i]);
    }
```

```
    return 0;
}
```

Ouput:

```
Enter the number of names: 5
Enter 5 names:
Ram
laxman
Shyam
Radhe
Abc

Sorted names:
Abc
Radhe
Ram
Shyam
laxman
harshitsingh@Harshits-MacBook-Air Question_6 % cd "/Users/harshitsingh/Documents/DA_CSE/Question_6/" && gcc ma
o main && "/Users/harshitsingh/Documents/DA_CSE/Question_6/"main
Enter the number of names: 2
Enter 2 names:
Harshit
Gajraj

Sorted names:
Gajraj
Harshit
```

Question 7:

```c
#include <stdio.h>

char *day_name(int n) {
    char *days[] = {"Sunday", "Monday", "Tuesday", "Wednesday", "Thursday",
"Friday", "Saturday"};
    n = (n % 7 + 7) % 7;
    return days[n];
}

int main() {
    int day_number;
    printf("Enter a number: ");
    scanf("%d", &day_number);
    printf("The day corresponding to %d is %s.\n", day_number,
day_name(day_number));
    return 0;
}
```

Ouput:

```
harshitsingh@Harshits-MacBook-Air Question_6 % cd "/Users/ha
o main && "/Users/harshitsingh/Documents/DA_CSE/Question_7/"
Enter a number: 5
The day corresponding to 5 is Friday.
harshitsingh@Harshits-MacBook-Air Question_7 % cd "/Users/ha
o main && "/Users/harshitsingh/Documents/DA_CSE/Question_7/"
Enter a number: 8
The day corresponding to 8 is Monday.
harshitsingh@Harshits-MacBook-Air Question_7 % cd "/Users/ha
o main && "/Users/harshitsingh/Documents/DA_CSE/Question_7/"
Enter a number: 9
The day corresponding to 9 is Tuesday.
harshitsingh@Harshits-MacBook-Air Question_7 %
```

Question 8:

```c
#include <stdio.h>
#include
<stdlib.h>
void add_matrices(int *mat1, int *mat2, int *result, int r, int c) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            // Accessing elements of matrices using pointer arithmetic
            *(result+ i * c + j) = *(mat1 + i * c + j) + *(mat2 + i * r + j);
        }
    }
}

// Function to display a matrix
void display_matrix(int *matrix, int r, int c) {
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            // Accessing elements of the matrix using pointer arithmetic
            printf("%d ", *(matrix + i * c + j));
        }
        printf("\n");
    }
}

int main() {
    int rows, cols;
    printf("Enter the number of rows and columns of the matrices: ");
    scanf("%d %d", &rows, &cols);
    int *mat1 = (int *)malloc(rows * cols * sizeof(int));
    int *mat2 = (int *)malloc(rows * cols * sizeof(int));
    int *result = (int *)malloc(rows * cols * sizeof(int));

    printf("Enter elements of matrix 1:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", mat1 + i * cols + j);
        }
    }

    printf("Enter elements of matrix 2:\n");
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            scanf("%d", mat2 + i * cols + j);
        }
    }

    add_matrices(mat1, mat2, result, rows, cols);

    // Displaying matrices
    printf("\nMatrix 1:\n");
```

```
    display_matrix(mat1, rows, cols);

    printf("\nMatrix 2:\n");
    display_matrix(mat2, rows, cols);

    printf("\nResultant Matrix:\n");
    display_matrix(result, rows, cols);

    free(mat1);
    free(mat2);
    free(result);

    return 0;
}
```

Ouput:

```
Enter the number of rows and columns of the matrices: 3 3
Enter elements of matrix 1:
1
2
3
4
5
6
7
8
9
Enter elements of matrix 2:
0
0
0
0
0
0
0
0
0

Matrix 1:
1 2 3
4 5 6
7 8 9

Matrix 2:
0 0 0
0 0 0
0 0 0

Resultant Matrix:
1 2 3
4 5 6
7 8 9
```

Question 9:

Question 9:

```c
#include <stdio.h>

float yardsFeetToMiles(int *yards, int *feet) {
    float total_feet=(*feet)+ (*yards * 3);
    return total_feet / 5280; // Convert feet to miles
}

int main() {
    int yards,feet;
    scanf("%d",&yards);
    scanf("%d",&feet);
    float miles1 = yardsFeetToMiles(&yards, &feet);

    printf("%d Yards and %d feet is %.2f miles.\n", yards,feet,miles1);

    return 0;
}
```

Output:

```
cd "/Users/harshitsingh/Documents/DA_CSE/Question_9/" && gcc m
/Question_9/"main
harshitsingh@Harshits-MacBook-Air DA_CSE % cd "/Users/harshits
ain && "/Users/harshitsingh/Documents/DA_CSE/Question_9/"main
1760
1320
1760 Yards and 1320 feet is 1.25 miles.
harshitsingh@Harshits-MacBook-Air Question_9 % cd "/Users/hars
-o main && "/Users/harshitsingh/Documents/DA_CSE/Question_9/"m
1000
5000
1000 Yards and 5000 feet is 1.52 miles.
harshitsingh@Harshits-MacBook-Air Question_9 %
```

Question 10:

```c
#include <stdio.h>
struct Item {
    float cost;
    int quantity;
};

// Function to calculate total inventory value
float calculate_inventory_value(struct Item items[], int num_items) {
    float total_value = 0.0;
    for (int i = 0; i < num_items; i++) {
        total_value += items[i].cost * items[i].quantity;
    }
    return total_value;
}

// Function to calculate total number of items in inventory
int calculate_total_items(struct Item items[], int num_items) {
    int total_items = 0;
    for (int i = 0; i < num_items; i++) {
        total_items += items[i].quantity;
    }
    return total_items;
}

int main() {
    struct Item inventory[15];
    int num_items = 0;
    printf("Enter information for each item (up to %d items):\n", 15);
    while (num_items < 15) {
        printf("Item %d: Cost and Quantity (separated by space): ", num_items + 1);
        if (scanf("%f %d", &inventory[num_items].cost,
&inventory[num_items].quantity) != 2)
            break;
        num_items++;
    }

    float total_value = calculate_inventory_value(inventory, num_items);
    int total_items = calculate_total_items(inventory, num_items);

    printf("\nTotal number of items in inventory: %d\n", total_items);
    printf("Total value of inventory: $%.2f\n", total_value);

    printf("\nEnter changes to the inventory:\n");
    for (int i = 0; i < num_items; i++) {
        printf("Change in quantity for item %d: ", i + 1);
        int change;
        scanf("%d", &change);
        inventory[i].quantity += change;
    }
```

```
    total_value = calculate_inventory_value(inventory, num_items);
    total_items = calculate_total_items(inventory, num_items);

    printf("\nUpdated total number of items in inventory: %d\n", total_items);
    printf("Updated total value of inventory: %.2f\n", total_value);

    return 0;
}
```

Output:

```
Item 1: Cost and Quantity (separated by space): 100 1
Item 2: Cost and Quantity (separated by space): 100 1
Item 3: Cost and Quantity (separated by space): 100 1
Item 4: Cost and Quantity (separated by space): 100 1
Item 5: Cost and Quantity (separated by space): 100 1
Item 6: Cost and Quantity (separated by space): 100 1
Item 7: Cost and Quantity (separated by space): 100 1
Item 8: Cost and Quantity (separated by space): 100 1
Item 9: Cost and Quantity (separated by space): 100 1
Item 10: Cost and Quantity (separated by space): 100 1
Item 11: Cost and Quantity (separated by space): 100 1
Item 12: Cost and Quantity (separated by space): 100 1
Item 13: Cost and Quantity (separated by space): 100 1
Item 14: Cost and Quantity (separated by space): 100 1
Item 15: Cost and Quantity (separated by space): 100 1

Total number of items in inventory: 15
Total value of inventory: $1500.00

Enter changes to the inventory:
Change in quantity for item 1: 2
Change in quantity for item 2: 2
Change in quantity for item 3: 2
Change in quantity for item 4: 2
Change in quantity for item 5: 2
Change in quantity for item 6: 2
Change in quantity for item 7: 2
Change in quantity for item 8: 2
Change in quantity for item 9: 2
Change in quantity for item 10: 2
Change in quantity for item 11: 2
Change in quantity for item 12: 2
Change in quantity for item 13: 2
Change in quantity for item 14: 2
Change in quantity for item 15: 2

Updated total value of inventory: 4500.00
harshitsingh@Harshits-MacBook-Air Question_10 %
```

Question 11:

```c
#include <stdio.h>

int main() {
    int total_money = 700;
    int total_coins;
    int twenty_five_paise, fifty_paise, one_rupee;
    total_coins = total_money * 100; // converting total money to total paise

    twenty_five_paise = total_coins / (25 + 50 + 100);
    fifty_paise = twenty_five_paise;
    one_rupee = twenty_five_paise;

    printf("Number of 25 paise coins: %d\n", twenty_five_paise);
    printf("Number of 50 paise coins: %d\n", fifty_paise);
    printf("Number of 1 rupee coins: %d\n", one_rupee);

    return 0;
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air Question_11 % cd "/Users/har
  -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_11/
Number of 25 paise coins: 400
Number of 50 paise coins: 400
Number of 1 rupee coins: 400
harshitsingh@Harshits-MacBook-Air Question_11 %
```

Question 12:

```c
#include <stdio.h>

int SPECIAL_NUMBER(int);
int main(){
    int a;
    scanf("%d",&a);
    return SPECIAL_NUMBER(a);

}
int SPECIAL_NUMBER(int a){
    int temp=a;
    int rev,digit;
    int last=a%10;
    int count=0;
    while(a!=0){
        a=a/10;
        count+=a%10;
    }
    if(last==count){
        printf("THIS IS A SPECIAL NUMBER");
    }
    else{
        printf("THIS IS NOT A SPECIAL NUMBER. ");
    }
    return 0;
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air Question_11 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_12/
246
THIS IS A SPECIAL NUMBER%
harshitsingh@Harshits-MacBook-Air Question_12 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_12/
729
THIS IS A SPECIAL NUMBER%
harshitsingh@Harshits-MacBook-Air Question_12 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_12/
264
THIS IS NOT A SPECIAL NUMBER. %
harshitsingh@Harshits-MacBook-Air Question_12 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_12/
381
THIS IS NOT A SPECIAL NUMBER. %
harshitsingh@Harshits-MacBook-Air Question_12 %
```

Question 13:

```c
#include <stdio.h>
int Bits(int n)
{
    int count = 0;
    while (n > 0) {
        count++;
        n &= (n - 1);
    }
    return count;
}

int FCount(int a, int b)
{
    return Bits(a ^ b);
}

int main()
{
    int a,b;
    scanf("%d %d",&a,&b);

    printf("%d\n", FCount(a, b));
    return 0;
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air Question_12 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_13/
7 10
3
harshitsingh@Harshits-MacBook-Air Question_13 % cd "/Users/har
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_13/
9 11
1
harshitsingh@Harshits-MacBook-Air Question_13 %
```

Question 14:

```c
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
void calculate_stats(float *weights, int n, float *sum, float *mean, float
*std_dev) {
    *sum = 0.0;
    for (int i = 0; i < n; i++) {
        *sum += *(weights + i);
    }
    *mean = *sum / n;
    float sum_of_squares = 0.0;
    for (int i = 0; i < n; i++) {
        sum_of_squares += (*(weights + i) - *mean) * (*(weights + i) - *mean);
    }
    *std_dev = sum_of_squares / n;
}

int main() {
    int n;
    printf("Enter the number of people: ");
    scanf("%d", &n);
    float *weights = (float *)malloc(n * sizeof(float));
    printf("Enter the weights of %d people:\n", n);
    for (int i = 0; i < n; i++) {
        scanf("%f", weights + i);
    }

    float sum, mean, std_dev;

    calculate_stats(weights, n, &sum, &mean, &std_dev);

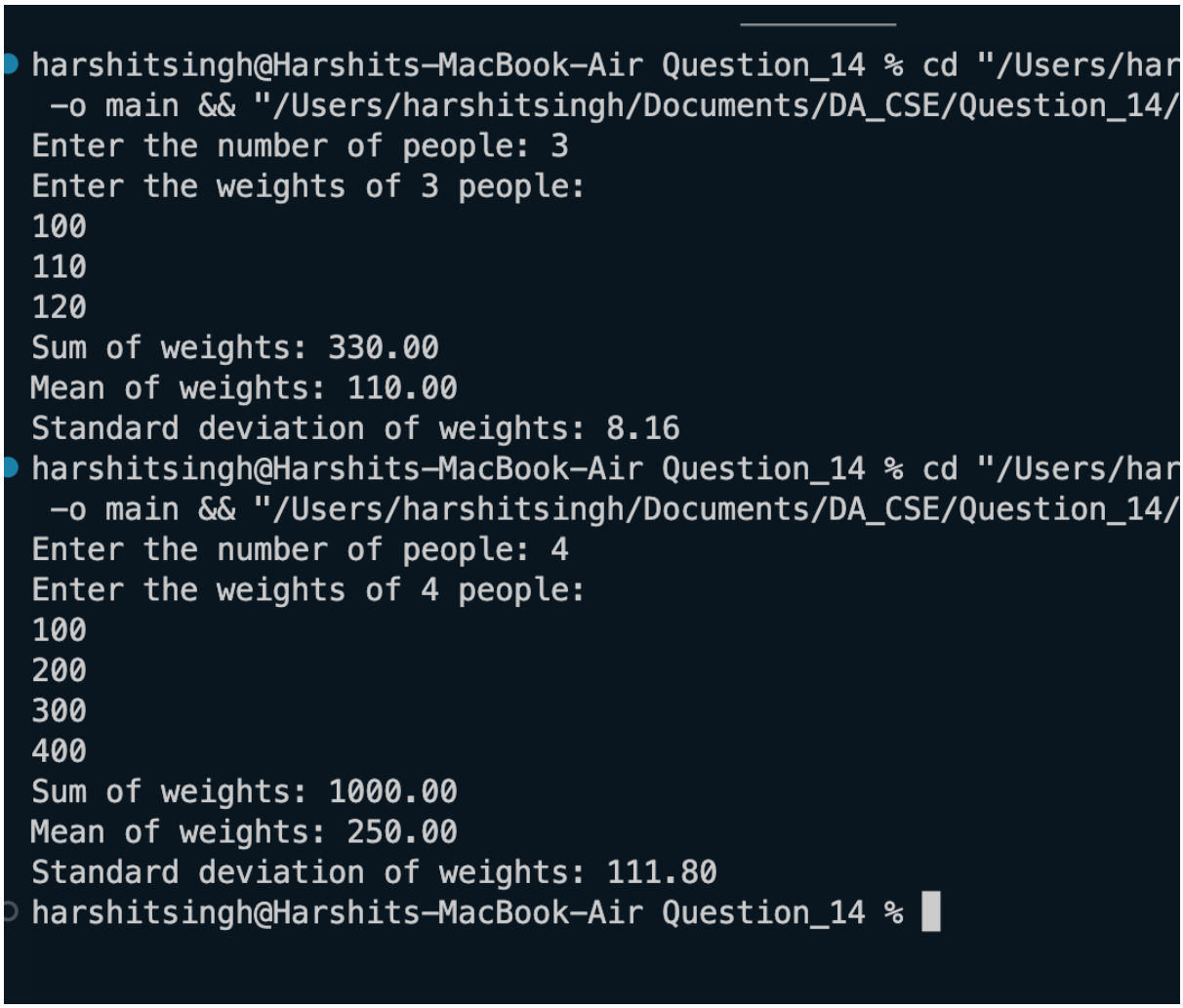

    std_dev = sqrt(std_dev);

    // Print results
    printf("Sum of weights: %.2f\n", sum);
    printf("Mean of weights: %.2f\n", mean);
    printf("Standard deviation of weights: %.2f\n", std_dev);

    // Free dynamically allocated memory
    free(weights);

    return 0;
}
```

Output:


harshitsingh@Harshits-MacBook-Air Question_14 % cd "/Users/har
-o main && "/Users/harshitsingh/Documents/DA_CSE/Question_14/
Enter the number of people: 3
Enter the weights of 3 people:
100
110
120
Sum of weights: 330.00
Mean of weights: 110.00
Standard deviation of weights: 8.16
harshitsingh@Harshits-MacBook-Air Question_14 % cd "/Users/har
-o main && "/Users/harshitsingh/Documents/DA_CSE/Question_14/
Enter the number of people: 4
Enter the weights of 4 people:
100
200
300
400
Sum of weights: 1000.00
Mean of weights: 250.00
Standard deviation of weights: 111.80
harshitsingh@Harshits-MacBook-Air Question_14 %

Question 15:

```c
#include <stdio.h>


int main() {
    int m, n;

    // Input the number of students (m) and the number of days (n)
    printf("Enter the number of students: ");
    scanf("%d", &m);
    printf("Enter the number of days in the month: ");
    scanf("%d", &n);

    // Data representing student attendance
    char data[31][10];

    // Input the attendance data for each day
    printf("Enter the attendance data for each day (P for present, A for
absent):\n");
    for (int i = 0; i < n; i++) {
        printf("Day %d: ", i + 1);
        for (int j = 0; j < m; j++) {
            scanf(" %c", &data[i][j]);
        }
    }

    // i. Find the maximum number of consecutive days on which all students were
present
    int max_present = 0;
    int present_count = 0;
    int start_present = -1;
    for (int i = 0; i < n; i++) {
        int all_present = 1; // Assume all students are present
        for (int j = 0; j < m; j++) {
            if (data[i][j] == 'A') {
                all_present = 0; // If any student is absent, set all_present to 0
                break;
            }
        }
        if (all_present) {
            present_count++;
            if (start_present == -1)
                start_present = i;
            if (present_count > max_present)
                max_present = present_count;
        } else {
            present_count = 0;
            start_present = -1;
        }
    }
```

```
    // ii. Find the maximum number of consecutive days on which all students were
absent
    int max_absent = 0;
    int absent_count = 0;
    int start_absent = -1;
    for (int i = 0; i < n; i++) {
        int all_absent = 1; // Assume all students are absent
        for (int j = 0; j < m; j++) {
            if (data[i][j] == 'P') {
                all_absent = 0; // If any student is present, set all_absent to 0
                break;
            }
        }
        if (all_absent) {
            absent_count++;
            if (start_absent == -1)
                start_absent = i;
            if (absent_count > max_absent)
                max_absent = absent_count;
        } else {
            absent_count = 0;
            start_absent = -1;
        }
    }

    // iii. Find the days in which the number of students present is greater than
the number of students absent
    printf("present highest\n");
    for (int i = 0; i < n; i++) {
        int present = 0, absent = 0;
        for (int j = 0; j < m; j++) {
            if (data[i][j] == 'P')
                present++;
            else
                absent++;
        }
        if (present > absent)
            printf("%d ", i + 1);
    }
    printf("\n");

    // iv. Find the days in which the number of students absent is greater than the
number of students present
    printf("absent highest\n");
    for (int i = 0; i < n; i++) {
        int present = 0, absent = 0;
        for (int j = 0; j < m; j++) {
            if (data[i][j] == 'P')
                present++;
            else
```

```
                absent++;
        }
        if (absent > present)
            printf("%d ", i + 1);
    }
    printf("\n");

    // v. Find the days in which the number of students present and absent is equal
    printf("present absent equal\n");
    for (int i = 0; i < n; i++) {
        int present = 0, absent = 0;
        for (int j = 0; j < m; j++) {
            if (data[i][j] == 'P')
                present++;
            else
                absent++;
        }
        if (present == absent)
            printf("%d ", i + 1);
    }
    printf("\n");

    // Displaying the maximum consecutive days where all students were present
    printf("consecutive days present\n");
    printf("%d\n", max_present);
    if (max_present > 0) {
        for(int i=0;i<m;i++){
            int count=0;
            for(int j=0;j<n;j++){
                if(data[i][j]== 'P'){
                    count+=1;
                }
            }
            if(count==m){
                printf("%d ",i+1);
            }
        }
    }

    printf("\nconsecutive days Absent\n");
    printf("%d\n", max_absent);
    if (max_absent > 0) {
        for(int i=0;i<m;i++){
            int count1=0;
            for(int j=0;j<n;j++){
                if(data[i][j]== 'A'){
                    count1+=1;
                }
            }
            if(count1==m){
                printf("%d ",i+1);
```

```
            }
        }
    }

    return 0;
}
```

Output:

```
harshitsingh@Harshits-MacBook-Air Question_15 % cd "/Users/hars
 -o main && "/Users/harshitsingh/Documents/DA_CSE/Question_15/"
Enter the number of students: 4
Enter the number of days in the month: 4
Enter the attendance data for each day (P for present, A for ab
Day 1: PPPP
Day 2: AAAA
Day 3: PAPA
Day 4: PAAA
present highest
1
absent highest
2 4
present absent equal
3
consecutive days present
1
1
consecutive days Absent
1
2 %
harshitsingh@Harshits-MacBook-Air Question_15 %
```